MSK Biochemistry — NBME Study Guide

Glycolysis · Glycogen metabolism & storage diseases · GAGs & carbohydrates · Collagen & elastin

Red = extra high yield (classic NBME) Black = core content *Italic blue = mnemonic*

1. Glucose uptake

Transporter	Location	Note
GLUT1	RBCs, brain, BBB, placenta	Insulin-independent, basal uptake
GLUT2	Liver, pancreatic β cells, kidney & intestine (basolateral)	High Km — glucose sensor, bidirectional
GLUT3	Neurons, placenta	High affinity
GLUT4	**Skeletal muscle, adipose, heart**	**Insulin-dependent (vesicles move to membrane); exercise also translocates GLUT4**
GLUT5	Intestine, sperm	Fructose

2. Glycolysis (cytosol; all cells)

- Glucose → 2 pyruvate; net **2 ATP + 2 NADH** (anaerobic: pyruvate → lactate via LDH regenerates NAD^+; RBCs have no mitochondria — glycolysis only).
- **Irreversible, regulated steps: hexokinase/glucokinase, PFK-1 (rate-limiting), pyruvate kinase.**

Enzyme	Activators	Inhibitors
Hexokinase (most tissues; low Km, high affinity)	—	**Glucose-6-phosphate**
Glucokinase (liver, β cells; high Km, high Vmax)	Insulin	F-6-P (via GK regulatory protein)
PFK-1 (rate-limiting)	**AMP, fructose-2,6-bisphosphate**	**ATP, citrate**
Pyruvate kinase	**Fructose-1,6-bisphosphate (feedforward)**	ATP, alanine, glucagon (PKA phosphorylation in liver)
Pyruvate dehydrogenase (mitochondria)	NAD^+, ADP, Ca^{2+} (exercising muscle), insulin	Acetyl-CoA, NADH, ATP

- **Fructose-2,6-BP (made by PFK-2)**: fed state/insulin ↑ PFK-2 activity → ↑ F-2,6-BP → ↑ glycolysis; fasting/glucagon → ↑ FBPase-2 → ↓ F-2,6-BP → gluconeogenesis.
- Substrate-level phosphorylation steps: **phosphoglycerate kinase** (1,3-BPG → 3-PG) and **pyruvate kinase** (PEP → pyruvate). NADH made at G3P dehydrogenase.
- **PDH complex cofactors: TPP (B1), lipoic acid, CoA (B5), FAD (B2), NAD^+ (B3)** *("Tender Loving Care For Nancy")*. Arsenic inhibits lipoic acid. Same cofactors for α-KG DH.
- **Pyruvate kinase deficiency**: most common glycolytic enzyme defect → hemolytic anemia (RBCs can't make ATP), echinocytes; ↑ 2,3-BPG.
- 2,3-BPG made in RBCs via the Rapoport–Luebering shunt → right shift of O_2 curve.
- Pyruvate fates: lactate (LDH; anaerobic), alanine (ALT), oxaloacetate (pyruvate carboxylase, biotin; gluconeogenesis), acetyl-CoA (PDH; TCA).
- **Cori cycle**: muscle/RBC lactate → liver → glucose (costs 4 net ATP in liver). Glucose–alanine cycle carries N to liver.
- **Arsenic** also inhibits glyceraldehyde-3-P dehydrogenase step (arsenate substitutes for Pi → no ATP from 1,3-BPG). **Fluoride** inhibits enolase (used in blood glucose tubes).
- Fructose: fructokinase → F-1-P → aldolase B (bypasses PFK-1 → fast metabolism). **Hereditary fructose intolerance (aldolase B)**: hypoglycemia, vomiting, liver failure after fruit/sucrose. Essential fructosuria (fructokinase) benign. Galactose: **classic galactosemia (GALT)** → cataracts, liver failure, E. coli sepsis; galactokinase deficiency → cataracts only.

3. Glycogen metabolism

- Stored in liver (~10% weight; maintains blood glucose) and muscle (~1–2%; fuels muscle only). **Muscle lacks glucose-6-phosphatase** → can't release free glucose.
- **Synthesis**: glucose-6-P → glucose-1-P (phosphoglucomutase) → **UDP-glucose** (UDP-glucose pyrophosphorylase) → glycogenin primer → **glycogen synthase (rate-limiting; α1→4)** → branching enzyme (α1→6, every ~8–10 residues).
- **Breakdown**: **glycogen phosphorylase (rate-limiting; needs PLP/B6)** releases glucose-1-P from α1→4 until 4 residues from a branch → **debranching enzyme** (4-α-glucanotransferase moves 3 residues; α-1,6-glucosidase releases the branch glucose as free glucose) → glucose-1-P → glucose-6-P → (liver) glucose-6-phosphatase → glucose.
- Lysosomal α-1,4-glucosidase (acid maltase) degrades a small fraction of glycogen in lysosomes.

Signal	Effect
Insulin (fed)	**Activates protein phosphatase 1 → dephosphorylates: synthase ON, phosphorylase OFF → glycogen synthesis**
Glucagon (liver), epinephrine (liver + muscle)	**cAMP → PKA → phosphorylase kinase → phosphorylates: phosphorylase ON, synthase OFF → glycogenolysis**
Muscle contraction	Ca^{2+}–calmodulin activates phosphorylase kinase; AMP allosterically activates muscle phosphorylase
Liver glucose	Glucose inhibits liver phosphorylase; G-6-P activates synthase

Phosphorylation = breakdown mode (glycogenolysis); dephosphorylation = storage mode.

Glycogen storage diseases — MUST KNOW

Type	Enzyme	Key findings
I — Von Gierke	**Glucose-6-phosphatase (Ia) / transporter (Ib)**	**Severe fasting hypoglycemia, ↑↑ glycogen in liver & kidney (hepatomegaly, renomegaly), lactic acidosis, hyperuricemia (gout), hyperlipidemia, doll-like face; can't respond to glucagon. Tx: frequent cornstarch feeds; avoid fructose/galactose.**
II — Pompe	**Lysosomal acid α-1,4-glucosidase (acid maltase)**	**Cardiomegaly, hypertrophic cardiomyopathy, hypotonia, macroglossia, early death (infantile); normal blood glucose. *"Pompe trashes the Pump (heart, liver, muscle)"*. Enzyme replacement.**
III — Cori	**Debranching enzyme (α-1,6-glucosidase)**	**Milder Von Gierke: hypoglycemia, hepatomegaly, ± myopathy, normal lactate; short outer chains (limit dextrin) accumulate; gluconeogenesis intact.**
IV — Andersen	Branching enzyme	Abnormal long unbranched glycogen → cirrhosis, liver failure, death in early childhood.
V — McArdle	**Muscle glycogen phosphorylase (myophosphorylase)**	**Exercise intolerance, painful cramps, myoglobinuria (red urine) after exercise, ↑ CK, flat venous lactate curve on ischemic forearm test, "second wind" phenomenon; normal blood glucose.**
VI — Hers	Liver glycogen phosphorylase	Mild hypoglycemia, hepatomegaly.
VII — Tarui	Muscle PFK-1	Like McArdle + hemolytic anemia.

"Very Poor Carbohydrate Metabolism" → Von Gierke (I), Pompe (II), Cori (III), McArdle (V). All AR.

4. GAGs, proteoglycans & carbohydrates in MSK

- GAGs = repeating disaccharides (amino sugar + uronic acid), **negatively charged** → attract Na^+ and water → **resist compression** (cartilage); proteoglycan = core protein + GAGs.
- **Aggrecan** (cartilage): chondroitin sulfate + keratan sulfate on a core protein, attached by link protein to **hyaluronan** → huge aggregates.
- **Hyaluronic acid = only non-sulfated GAG, not bound to protein; synovial fluid viscosity, vitreous.** Heparin = only intracellular GAG (mast cells). Chondroitin sulfate most abundant (cartilage, bone). Keratan sulfate (cornea, cartilage). Dermatan sulfate (skin, vessels, valves). Heparan sulfate (basement membranes).

- Synthesis: core protein in RER → GAG chains added and sulfated in Golgi (PAPS sulfate donor). Degradation in lysosomes.
- **Mucopolysaccharidoses** (lysosomal GAG degradation defects; urinary GAGs, coarse facies, skeletal dysplasia): **Hurler** (α-L-iduronidase, AR) — corneal clouding, gargoylism, intellectual disability, early death; **Hunter** (iduronate sulfatase, X-linked) — no corneal clouding, aggressive behavior *("Hunters see clearly and aim for the X")*; heparan + dermatan sulfate accumulate.
- Osteoarthritis: ↓ proteoglycans → cartilage loses water-holding capacity; glucosamine/chondroitin supplements.

5. Collagen

- Structure: **triple helix of α chains with Gly-X-Y repeats (glycine every 3rd residue — only AA small enough for the center); X/Y often proline and hydroxyproline/hydroxylysine.**

Step	Location	Detail / disease
1. Translation of preprocollagen	RER	Signal peptide cleaved
2. Hydroxylation of Pro & Lys	**RER**	**Prolyl & lysyl hydroxylase need vitamin C (+ Fe^{2+}, O_2, α-KG) → scurvy**
3. Glycosylation of hydroxylysine	RER	Galactose/glucose added
4. Triple helix (procollagen) via disulfides at C-propeptides	**RER**	**Osteogenesis imperfecta — can't form triple helix (Gly substitutions)**
5. Secretion	Golgi → exocytosis	—
6. Propeptide cleavage → tropocollagen	**Extracellular**	**Procollagen peptidases; defect → Ehlers–Danlos (dermatosparaxis)**
7. Cross-linking → fibrils	**Extracellular**	**Lysyl oxidase (copper) covalent cross-links; Menkes disease (ATP7A, ↓ Cu: kinky brittle hair, hypotonia, growth delay); Ehlers–Danlos; lathyrism**

Disease	Defect	Features
Scurvy	**Vitamin C → ↓ hydroxylation**	**Swollen bleeding gums, perifollicular hemorrhage, corkscrew hairs, poor wound healing, hemarthrosis**
Osteogenesis imperfecta	**Type I collagen (COL1A1/2), AD**	**Multiple fractures with minimal trauma (mimics child abuse), blue sclerae, hearing loss (otosclerosis), dentinogenesis imperfecta**
Ehlers–Danlos	**Collagen processing (classic: type V; vascular: type III)**	**Hyperextensible skin, hypermobile joints, easy bruising; vascular type → arterial/organ rupture, berry aneurysms**
Menkes	Copper transport (ATP7A), XR	↓ lysyl oxidase → brittle kinky hair, neurodegeneration
Alport	Type IV collagen	Hematuria/nephritis, sensorineural deafness, lens/eye defects *("can't see, can't pee, can't hear")*

Collagen types: I Bone/skin/tendon; II Cartilage; III Reticulin (vessels, granulation); IV Basement membrane.

6. Elastin

- Stretchy protein in lungs, large arteries, elastic ligaments (ligamentum flavum), skin, vocal cords.
- **Rich in nonhydroxylated proline, glycine and lysine; lysyl oxidase forms desmosine cross-links**; assembled on a **fibrillin** scaffold. Broken down by elastase (inhibited by α_1-antitrypsin).
- **Marfan syndrome**: FBN1 (fibrillin-1), AD — tall, long limbs, arachnodactyly, pectus, **upward lens subluxation**, MVP, **aortic root dilation → dissection**, hypermobility. (Homocystinuria: downward lens dislocation, thrombosis, intellectual disability.)
- **α_1-antitrypsin deficiency**: unopposed elastase → panacinar emphysema (lower lobes, young, nonsmoker) + cirrhosis (misfolded PiZ retained in hepatocyte ER, PAS+ globules).
- Wrinkles of aging: ↓ collagen & elastin synthesis, UV damage (solar elastosis).

7. Rapid-fire NBME pearls

- **PFK-1 rate-limiting (activated by AMP, F-2,6-BP; inhibited by ATP, citrate); pyruvate kinase deficiency → hemolysis.**
- **Glycogen synthase / phosphorylase are rate-limiting; insulin dephosphorylates (store), glucagon/epi phosphorylate (break down).**
- **Von Gierke: hypoglycemia + lactic acidosis + gout + hepatomegaly. Pompe: cardiomegaly (lysosomal). Cori: Von Gierke-like but normal lactate. McArdle: cramps, myoglobinuria, flat lactate.**
- **Vitamin C → hydroxylation (scurvy); copper → lysyl oxidase cross-links (Menkes); OI = type I collagen, blue sclerae.**
- **Marfan = fibrillin-1 (upward lens, aortic root); hyaluronic acid non-sulfated; Hunter = X-linked, no corneal clouding.**